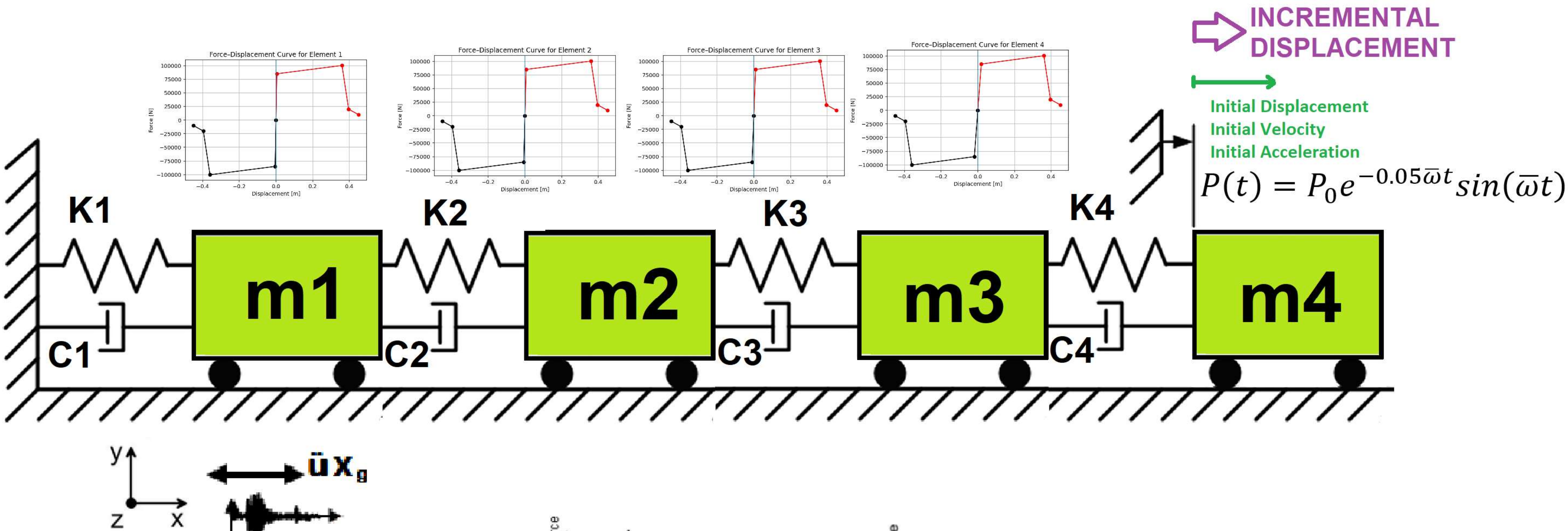
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

DETERMINATION OF OPTIMUM STRUCTURAL DUCTILITY RATIO FROM ENERGY DISSIPATION CAPACITY INDEX USING FINITE-DIFFERENCE NEWTON ITERATION AND OPENSEES

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)



$$Structural\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

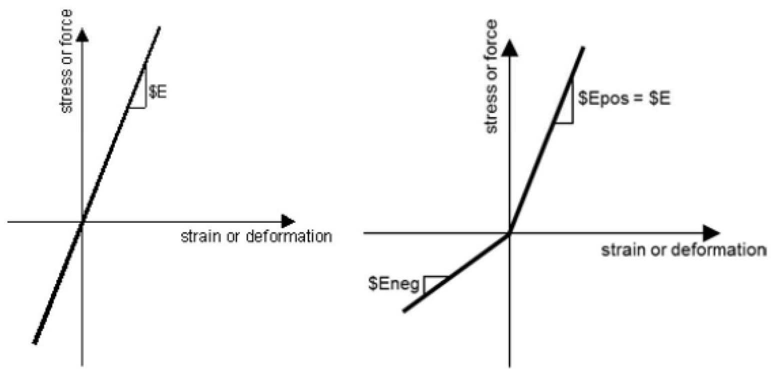
$$\Delta_d = Lateral\ Displaement\ from\ Dynamic\ Analysis$$

$$\Delta_y = Lateral\ Yield\ Displaement\ from\ Pushover\ Analysis$$

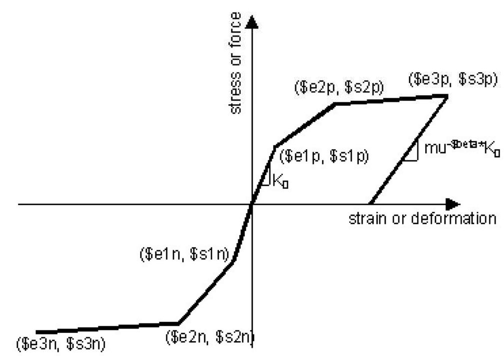
$$\Delta_u = Lateral\ Ultimate\ Displaement\ from\ Pushover\ Analysis$$

$$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$

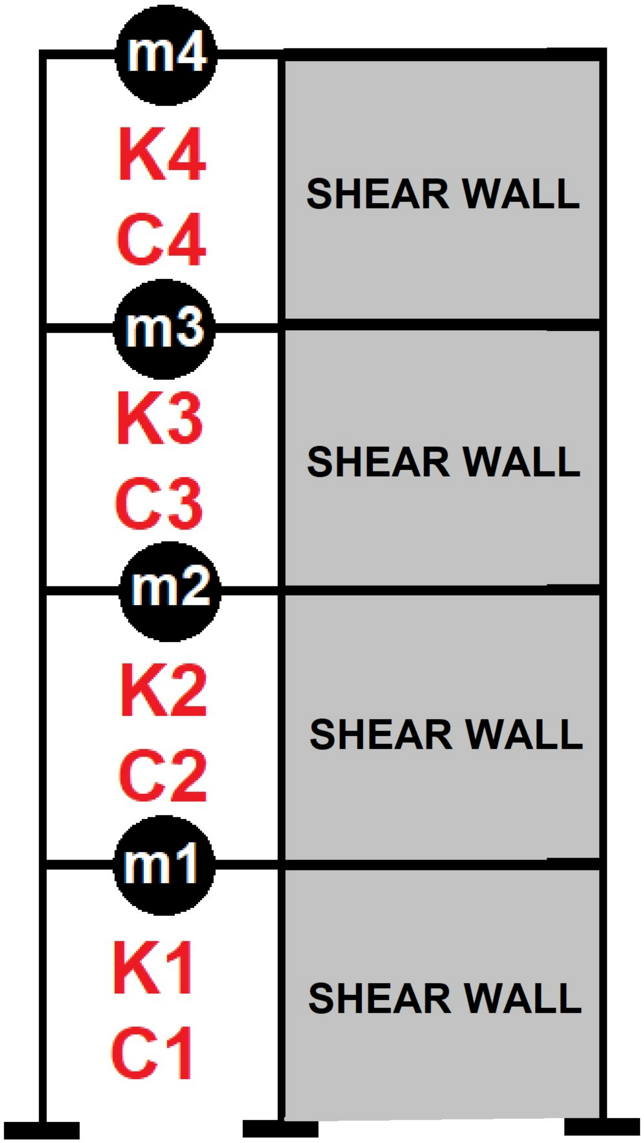


ELASTIC MATERIAL

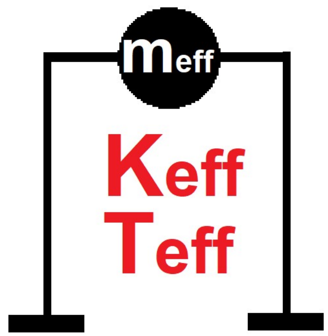


INELASTIC MATERIAL

Displacement-based seismic analysis transformation, converting a multi-degree-of-freedom (MDOF) system into an equivalent single-degree-of-freedom (SDOF) system for seismic assessment

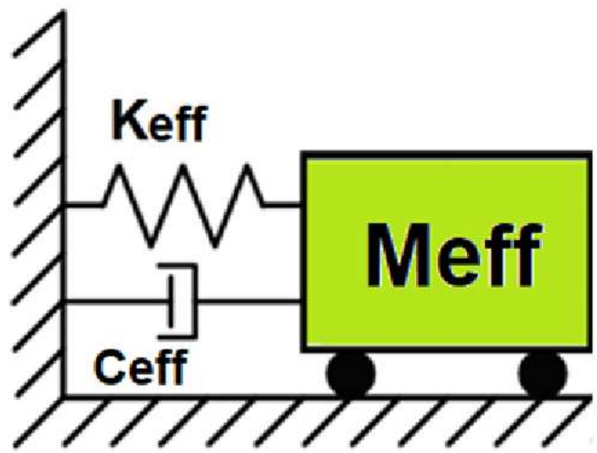


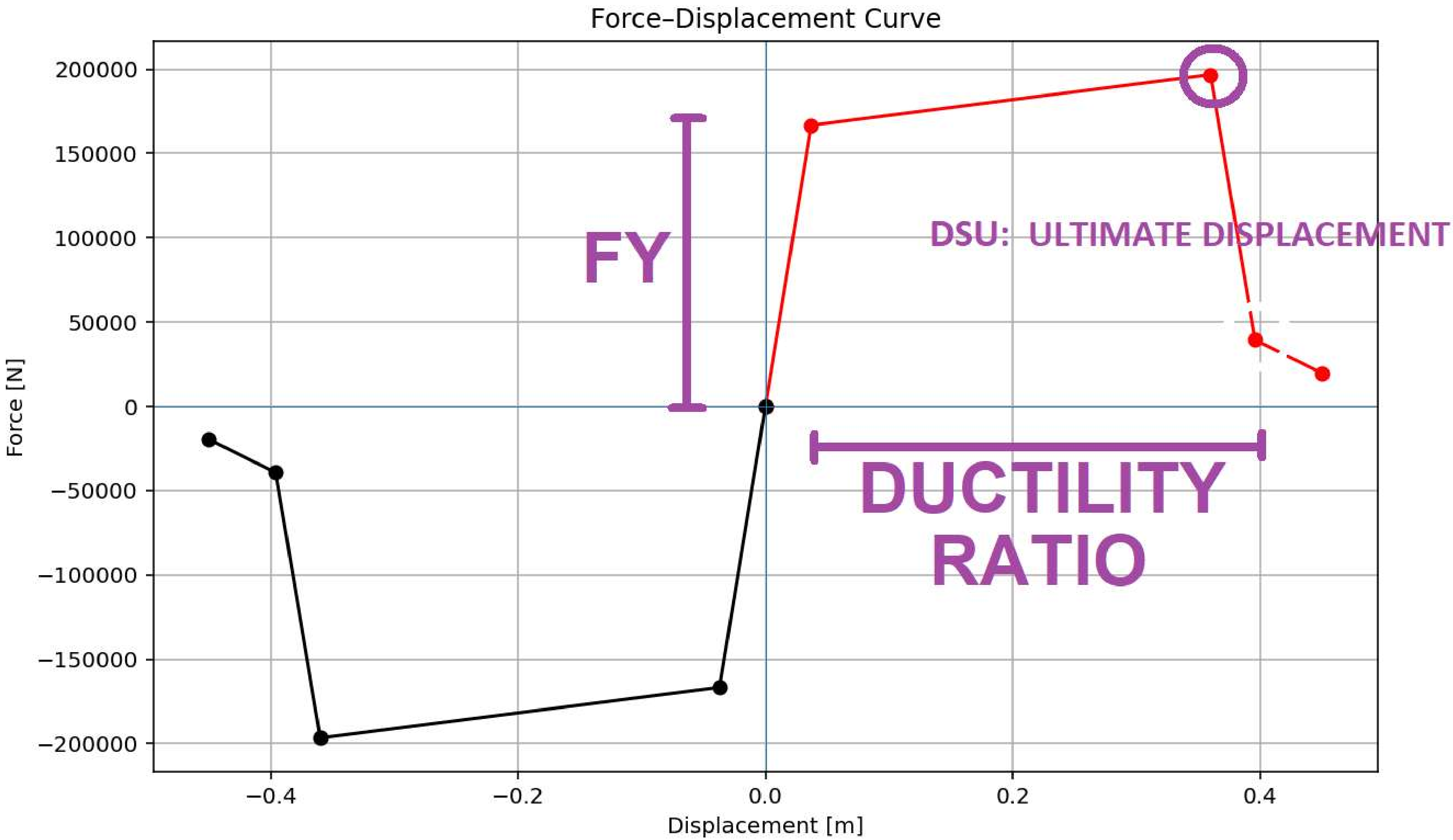
$\cong$



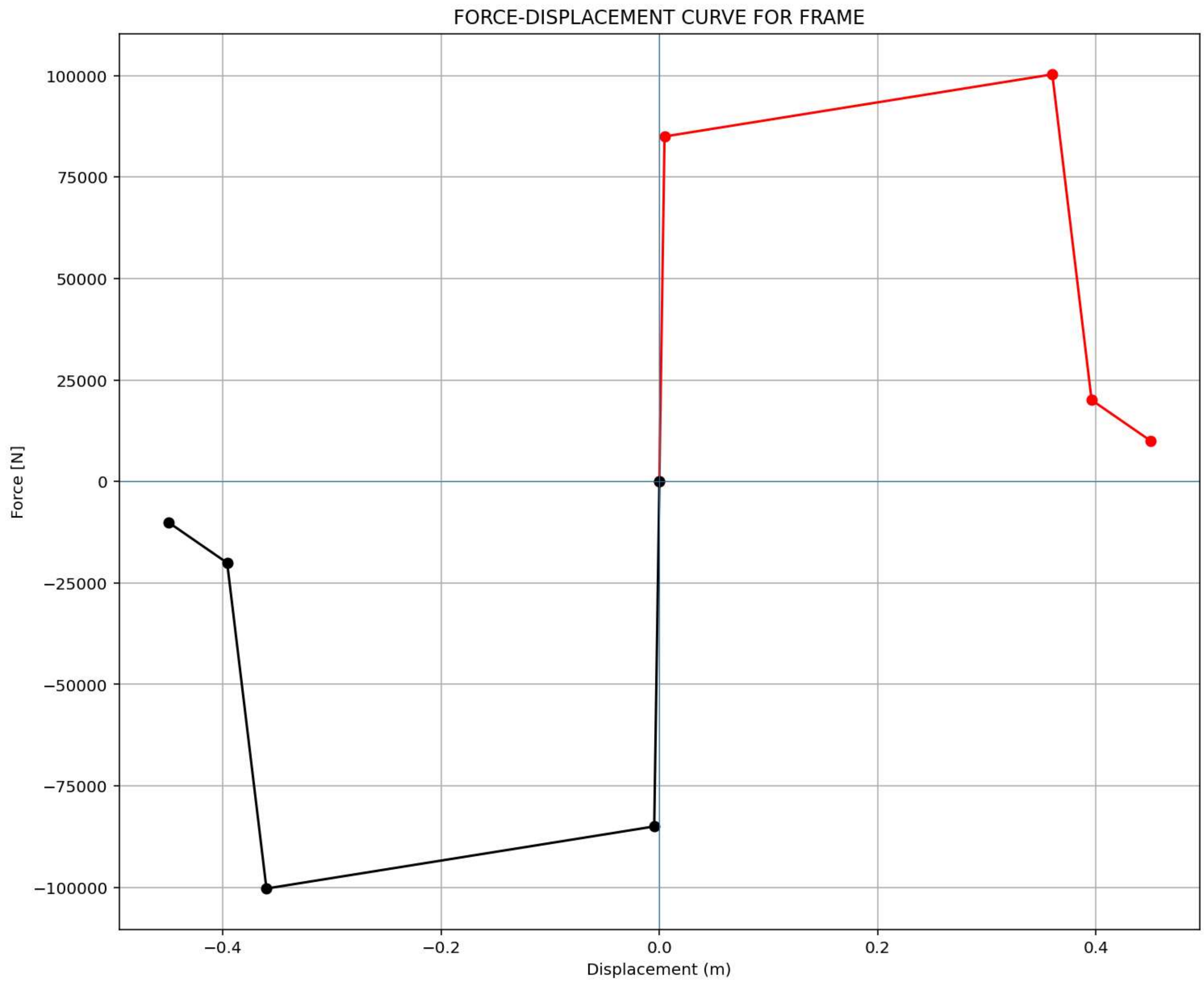
$$\Delta_d = \frac{\sum_{i=1}^{n} m_i \Delta_i^2}{\sum_{i=1}^{n} m_i \Delta_i} \qquad m_{eff} = \frac{\sum_{i=1}^{n} m_i \Delta_i}{\Delta_d}$$

$$k_{eff} = \frac{\sum_{i=1}^{n} k_i \Delta_i}{\Delta_d} \qquad T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$

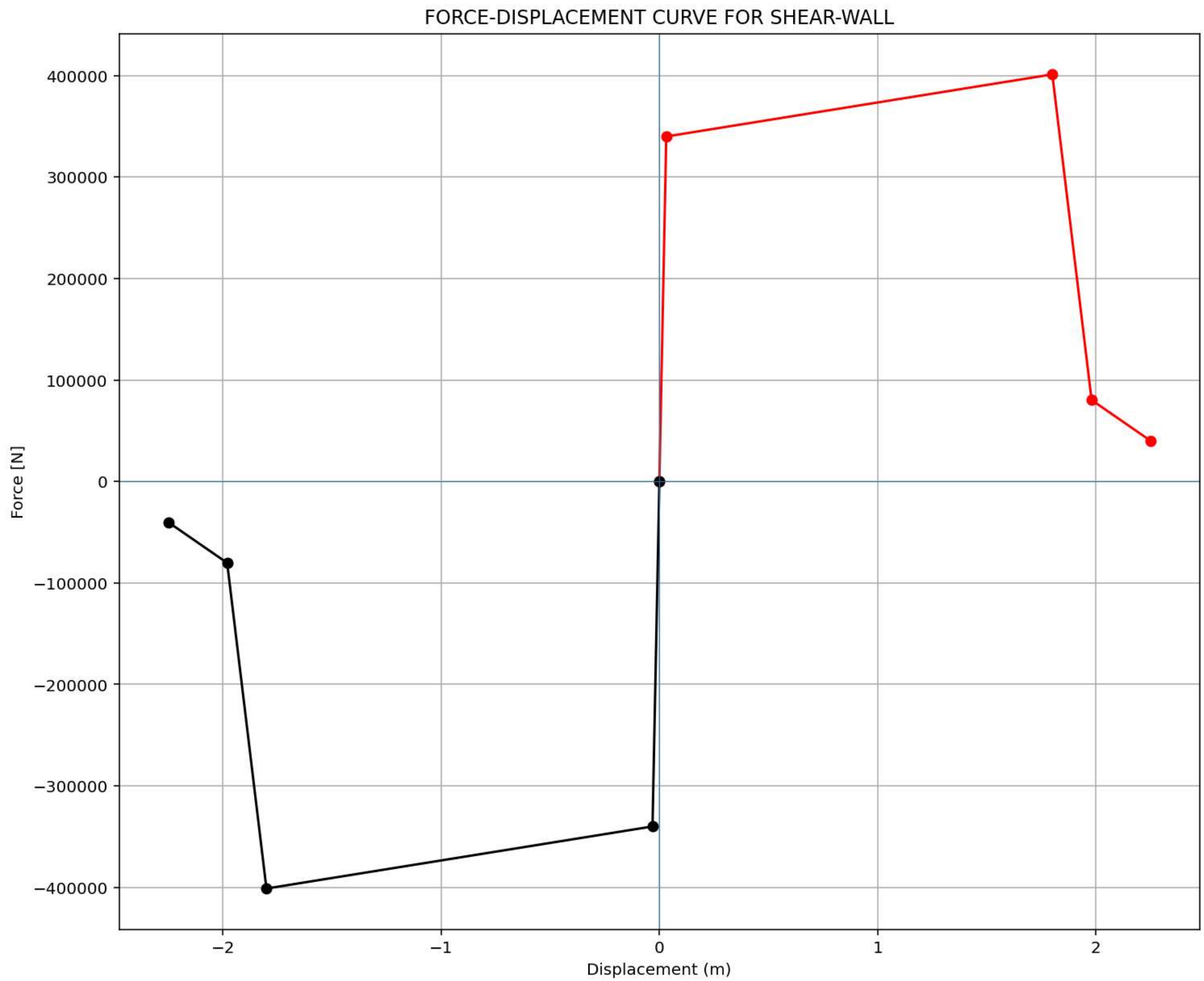


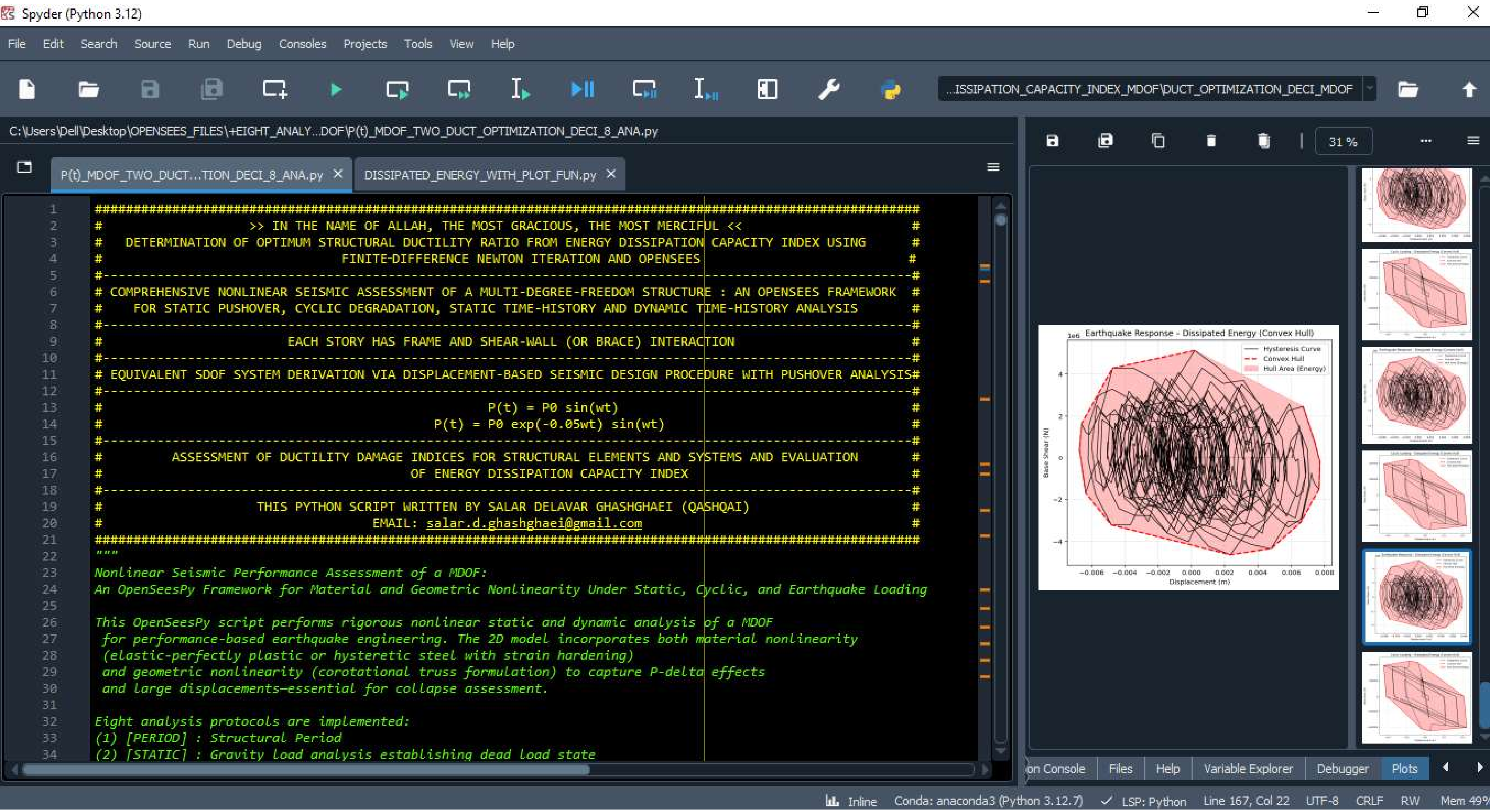


Force–Displacement Curve
200000
150000
100000
50000
0
−50000
−100000
−150000
−200000
Force [N]
−0.4
−0.2
0.0
0.2
0.4
Displacement [m]
FY
DSU: ULTIMATE DISPLACEMENT
DUCTILITY
RATIO



FORCE-DISPLACEMENT CURVE FOR FRAME
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
Force [N]
−0.4
−0.2
0.0
0.2
0.4
Displacement (m)


FORCE-DISPLACEMENT CURVE FOR SHEAR-WALL
400000
300000
200000
100000
0
−100000
−200000
−300000
−400000
Force [N]
−2
−1
0
1
2
Displacement (m)


Spyder (Python 3.12)
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DISSIPATED_ENERGY_WITH_PLOT_FUN.py
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
DETERMINATION OF OPTIMUM STRUCTURAL DUCTILITY RATIO FROM ENERGY DISSIPATION CAPACITY INDEX USING
FINITE-DIFFERENCE NEWTON ITERATION AND OPENSEES
COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A MULTI-DEGREE-FREEDOM STRUCTURE : AN OPENSEES FRAMEWORK
FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS
EACH STORY HAS FRAME AND SHEAR-WALL (OR BRACE) INTERACTION
EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED SEISMIC DESIGN PROCEDURE WITH PUSHOVER ANALYSIS#
P(t) = P0 sin(wt)
P(t) = P0 exp(-0.05wt) sin(wt)
ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND EVALUATION
OF ENERGY DISSIPATION CAPACITY INDEX
THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
EMAIL: salar.d.ghashghaei@gmail.com
"""
Nonlinear Seismic Performance Assessment of a MDOF:
An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and Earthquake Loading
This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a MDOF
for performance-based earthquake engineering. The 2D model incorporates both material nonlinearity
(elastic-perfectly plastic or hysteretic steel with strain hardening)
and geometric nonlinearity (corotational truss formulation) to capture P-delta effects
and large displacements—essential for collapse assessment.
Eight analysis protocols are implemented:
(1) [PERIOD] : Structural Period
(2) [STATIC] : Gravity load analysis establishing dead load state
31 %
Earthquake Response - Dissipated Energy (Convex Hull)
Hysteresis Curve
Convex Hull
Hull Area (Energy)
Base Shear (N)
Displacement (m)
on Console Files Help Variable Explorer Debugger Plots
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 167, Col 22 UTF-8 CRLF RW Mem 49%

Spyder (Python 3.12)

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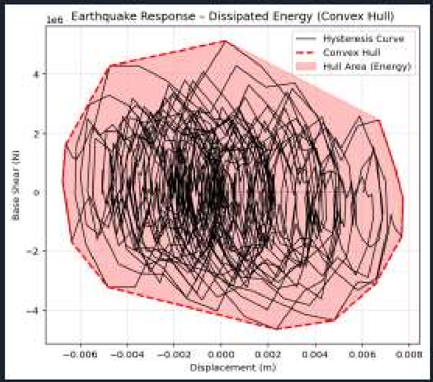
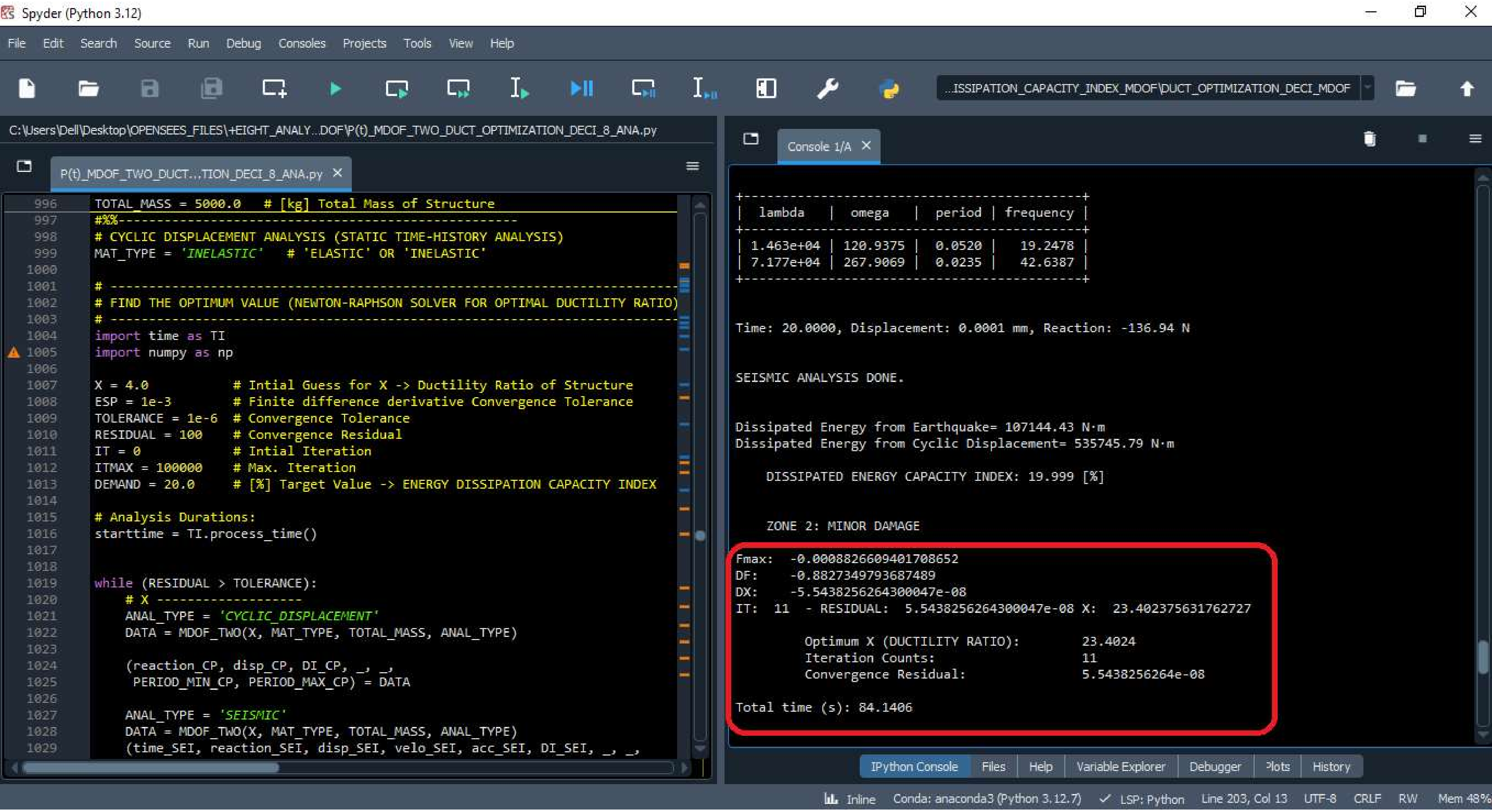
P(t)_MDOF_TWO_DUCT...TION_DECI_8_ANA.py | DISSIPATED_ENERGY_WITH_PLOT_FUN.py

```python
# EVALUATION OF DISSIPATED ENERGY CAPACITY INDEX
# THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
def DISSIPATED_ENERGY_FUN_WITH_PLOT(displacement, base_shear, title="Hysteresis Curve"):
    """
    Compute dissipated energy using convex hull and plot the hysteresis curve
    with the outer hull area shaded.

    Parameters
    ----------
    displacement : array-like
    base_shear   : array-like
    title        : str

    Returns
    -------
    float
        Area of convex hull (dissipated energy)
    """
    import numpy as np
    from scipy.spatial import ConvexHull
    import matplotlib.pyplot as plt
    displacement = np.asarray(displacement)
    base_shear  = np.asarray(base_shear)

    if displacement.size != base_shear.size:
        raise ValueError("Displacement and base shear arrays must have equal lengths.")

    points = np.column_stack((displacement, base_shear))
    hull = ConvexHull(points)
    area = hull.volume   # 2D hull → area

    fig, ax = plt.subplots(figsize=(7, 6))

    # Plot full hysteresis
    ax.plot(displacement, base_shear, 'k-', linewidth=1, label="Hysteresis Curve")
```



Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 1, Col 1 | UTF-8 | CRLF | RW | Mem 49%

OPTIMIZATION PROBLEM FOR ENERGY DISSIPATION CAPACITY INDEX (EDCI)


Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
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P(t)_MDOF_TWO_DUCT...TION_DECI_8_ANA.py
TOTAL_MASS = 5000.0   # [kg] Total Mass of Structure
#%%--
# CYCLIC DISPLACEMENT ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
# ---
# FIND THE OPTIMUM VALUE (NEWTON-RAPHSON SOLVER FOR OPTIMAL DUCTILITY RATIO)
# ---
import time as TI
import numpy as np
X = 4.0   # Intial Guess for X -> Ductility Ratio of Structure
ESP = 1e-3   # Finite difference derivative Convergence Tolerance
TOLERANCE = 1e-6 # Convergence Tolerance
RESIDUAL = 100   # Convergence Residual
IT = 0   # Intial Iteration
ITMAX = 100000   # Max. Iteration
DEMAND = 20.0   # [%] Target Value -> ENERGY DISSIPATION CAPACITY INDEX
# Analysis Durations:
starttime = TI.process_time()
while (RESIDUAL > TOLERANCE):
    # X ------------------
    ANAL_TYPE = 'CYCLIC_DISPLACEMENT'
    DATA = MDOF_TWO(X, MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
    (reaction_CP, disp_CP, DI_CP, _, _,
     PERIOD_MIN_CP, PERIOD_MAX_CP) = DATA
    ANAL_TYPE = 'SEISMIC'
    DATA = MDOF_TWO(X, MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
    (time_SEI, reaction_SEI, disp_SEI, velo_SEI, acc_SEI, DI_SEI, _, _,
Console 1/A
lambda	omega	period	frequency
1.463e+04	120.9375	0.0520	19.2478
7.177e+04	267.9069	0.0235	42.6387
Time: 20.0000, Displacement: 0.0001 mm, Reaction: -136.94 N
SEISMIC ANALYSIS DONE.
Dissipated Energy from Earthquake= 107144.43 N·m
Dissipated Energy from Cyclic Displacement= 535745.79 N·m
DISSIPATED ENERGY CAPACITY INDEX: 19.999 [%]
ZONE 2: MINOR DAMAGE
Fmax: -0.0008826609401708652
DF: -0.8827349793687489
DX: -5.5438256264300047e-08
IT: 11 - RESIDUAL: 5.5438256264300047e-08 X: 23.402375631762727
Optimum X (DUCTILITY RATIO): 23.4024
Iteration Counts: 11
Convergence Residual: 5.5438256264e-08
Total time (s): 84.1406
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 203, Col 13 UTF-8 CRLF RW Mem 48%

Dissipated Energy from Earthquake= 107144.43 N·m
Dissipated Energy from Cyclic Displacement= 535745.79 N·m

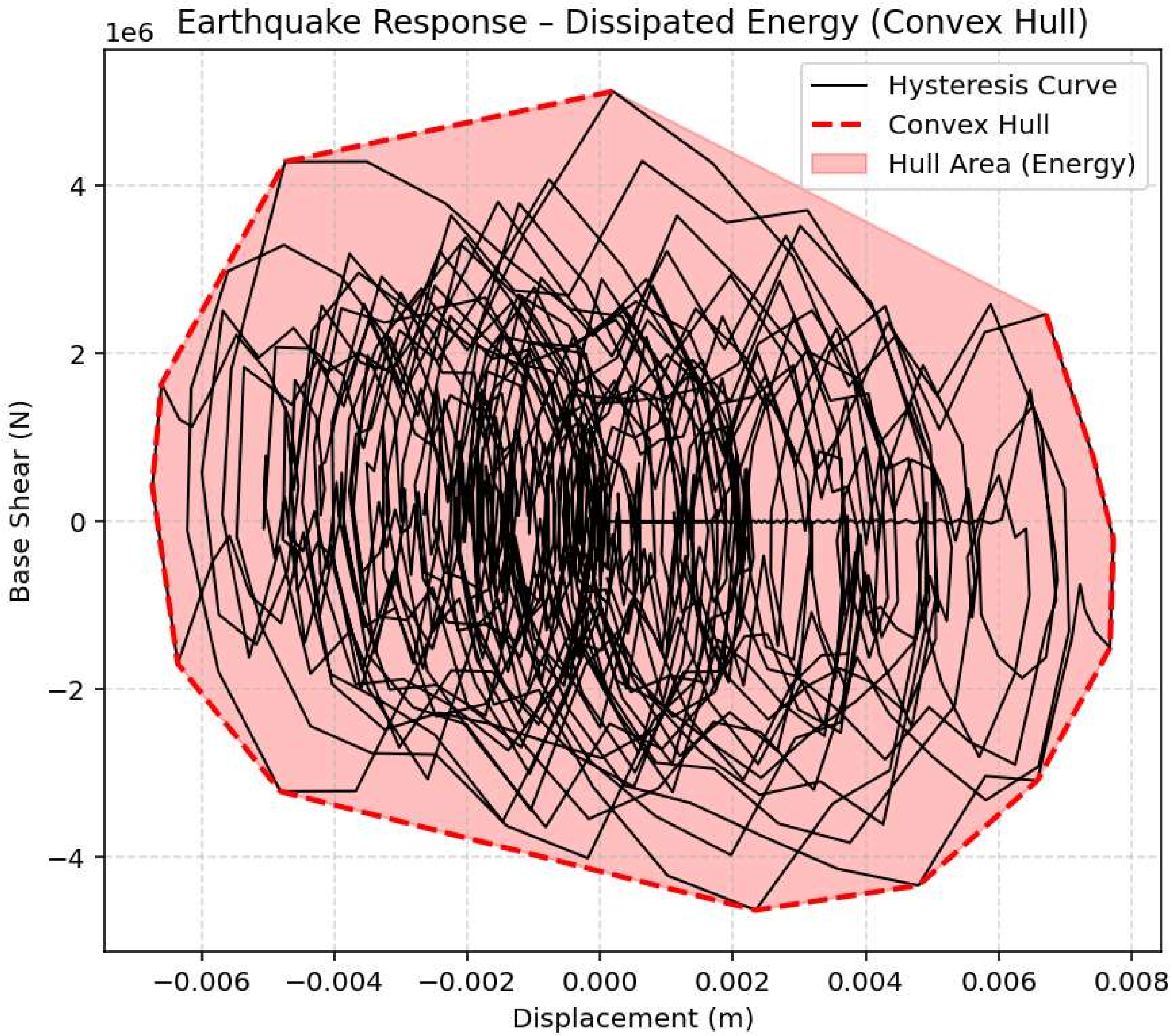
DISSIPATED ENERGY CAPACITY INDEX: 19.999 [%]

ZONE 2: MINOR DAMAGE

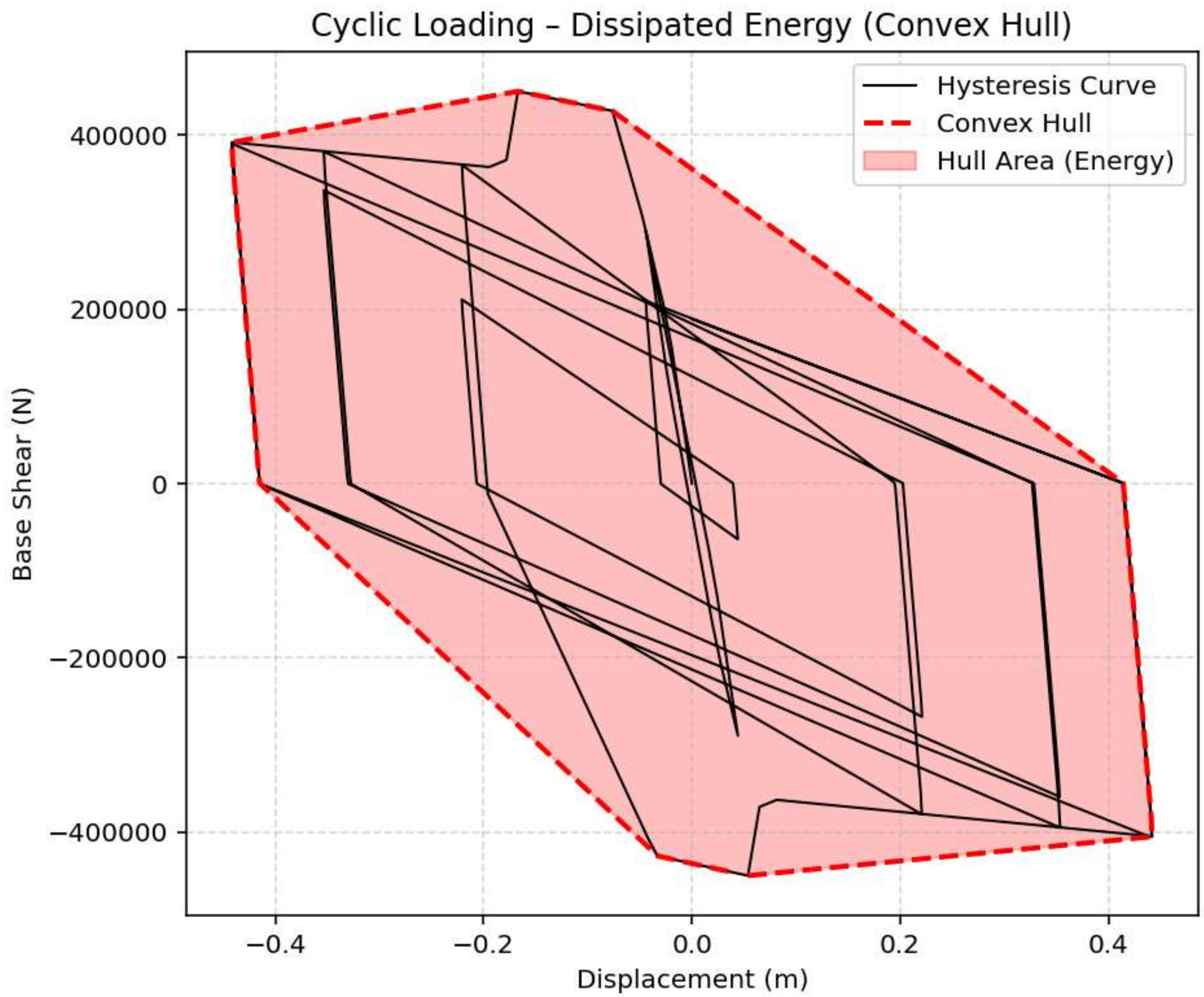
Fmax: -0.0008826609401708652
DF: -0.8827349793687489
DX: -5.5438256264300047e-08
IT: 11 - RESIDUAL: 5.5438256264300047e-08 X: 23.402375631762727

Optimum X (DUCTILITY RATIO): 23.4024
Iteration Counts: 11
Convergence Residual: 5.5438256264e-08

Total time (s): 84.1406



Earthquake Response – Dissipated Energy (Convex Hull)
1e6
Hysteresis Curve
Convex Hull
Hull Area (Energy)
Base Shear (N)
Displacement (m)
4
2
0
−2
−4
−0.006
−0.004
−0.002
0.000
0.002
0.004
0.006
0.008


Cyclic Loading – Dissipated Energy (Convex Hull)
Hysteresis Curve
Convex Hull
Hull Area (Energy)
Base Shear (N)
400000
200000
0
−200000
−400000
−0.4
−0.2
0.0
0.2
0.4
Displacement (m)